
MRQ : Support Multiple Quantization Schemes through Model Re-Quantization

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Abstract

Despite the proliferation of diverse hardware accelerators (e.g., NPU, TPU, DPU), deploying deep learning models on edge devices with fixed-point hardware is still challenging due to complex model quantization and conversion. Existing model quantization frameworks like Tensorflow QAT [1], TFLite PTQ [2], and Qualcomm AIMET [3] supports only a limited set of quantization schemes (e.g., only asymmetric per-tensor quantization in TF1.x QAT [4]). Accordingly, deep learning models cannot be easily quantized for diverse fixed-point hardwares, mainly due to slightly different quantization requirements. In this paper, we envision a new type of model quantization approach called MRQ (model re-quantization), which takes existing quantized models and quickly transforms the models to meet different quantization requirements (e.g., asymmetric $\rightarrow$ symmetric, non-power-of-2 scale $\rightarrow$ power-of-2 scale). Re-quantization is much simpler than quantizing from scratch because it avoids costly re-training and provides support for multiple quantization schemes simultaneously. To minimize re-quantization error, we developed a new set of re-quantization algorithms including weight correction and rounding error folding. We have demonstrated that MobileNetV2 QAT model [7] can be quickly re-quantized into two different quantization schemes (i.e., symmetric and symmetric+power-of-2 scale) with less than 0.64 units of accuracy loss. We believe our work is the first to leverage this concept of re-quantization for model quantization and models obtained from the re-quantization process have been successfully deployed on NNA in the Echo Show devices.

1 Introduction

With the popularity of deep neural networks across various use-cases (e.g., CV/AR, Audio/Speech, NLP), there is an increasing demand for methods that make deep networks run efficiently on resource-limited edge devices. One of most compelling methods is model quantization, where the float 32-bit weights and activations are quantized to 8-bit integer. Then, the quantized models can be run efficiently on fixed-point hardware with several benefits (i.e., 2-3x faster inference, 4x less memory and bandwidth usage, less chip BOM cost and energy consumption).

There are many different types of quantization schemes for deep learning models (e.g., per-tensor vs. per-channel, symmetric vs. asymmetric, power-of-2 scale vs. non-power-of-2 scale) (refer to related work for details). Depending upon the specific hardware configuration and relative importance of

Preprint. Under review.